

Technology-assisted Risk of Bias Assessment in RCTs using RobotReviewer: An Evaluation Study

Julian Hirt ^{1,2}, Jasmin Meichlinger ¹, Petra Schumacher ³, Gerhard Mueller ³

¹ Institute for Applied Nursing Science, University for Applied Sciences FHS St.Gallen, Switzerland

² Institute for Health and Nursing Science, Medical Faculty, Martin Luther University Halle-Wittenberg, Halle (Saale), Germany

³ Institute for Nursing Science, Department for Nursing Science and Gerontology, UMIT – Health & Life Sciences University, Hall in Tyrol, Austria

Background and aim

Online tools may support researchers in conducting systematic reviews [1]. RobotReviewer enables an automated risk of bias assessment in RCTs using machine learning [2]. An evaluation study [3] including 1180 trials from different health topics, demonstrated a Cohen’s kappa agreement between .10 and .48 between RobotReviewer and human assessment in different risk of bias domains. Furthermore, the analysis yielded a sensitivity between .28 and .76 and a specificity between .72 and .90 for detecting a low risk of bias in different domains. Since the sample of trials was not nursing-specific, it is unclear whether risk of bias can be assessed with the RobotReviewer in nursing-related RCTs. Therefore, the aim of this study was to evaluate the reliability of RobotReviewer’s risk of bias assessment in nursing-related RCTs.

Method

Research design: Evaluation study

- Index test: Risk of bias assessment of nursing-related RCTs via RobotReviewer
- Reference test: Risk of bias assessment of nursing-related RCTs reported in Cochrane reviews

Literature search process

Cochrane reviews with nurs* in title were identified in MEDLINE via PubMed on August 30th, 2018 (Figure 1)

Inclusion criteria

Electronical availability of full text of RCTs and English language of RCTs

Data extraction and assessment

Two independent research teams

Results

The selection process yielded 190 RCTs published between 1959 and 2016 in 23 Cochrane reviews published between 2000 and 2018 (Figure 1). Missing assessments of risk of bias domains in Cochrane reviews or RobotReviewer yielded varying sample sizes per risk of bias domain. Cohen’s Kappa were moderate for randomization (.52), allocation concealment (.60), and for blinding of personal/patients (.43). Blinding of outcome assessors had only slight agreement (.04). Percentage observed agreement was in all domains ≥.50. Sensitivity ranged from .44 to .88 and specificity from .48 to .95. Positive predictive value was highest for allocation concealment (.79) and lowest for blinding assessors (.25) (Tables 1 and 2).

Figure 1: Flowchart of literature search process

